THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(a)** Both A and R are true and R is the correct explanation of A.	H^+ is needed for acidic behaviour.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene prevents contact with air/water.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The properties of the convex mirror explain its use.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
11	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
12	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The energy left is too little to support further levels.
14	**(d)** 44 g	Mass is conserved: 12 + 32 = 44 g.
15	**(d)** 7	Complete neutralisation to NaCl.
16	**(a)** Y	Reacting with cold water indicates more reactivity.
17	**(a)** Both A and R are true and R is the correct explanation of A.	$2\,Mg + O_2 \longrightarrow 2\,MgO$: two reactants give one product, so R explains A.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Without water the acid does not ionise, so no H^+ is available to change the litmus; R explains A.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene cuts off air and moisture, preventing the violent reaction.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Self-linking into long chains, branches and rings (catenation) together with four bonds per atom gives a huge number of compounds.

SECTION — B (Very Short Answer) — Answers 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Mass of CO_2 = 12 + 32 = 44 g (1); Law of conservation of mass (1).	1+1
22	CO_2 (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$, lime water turns milky (1).	1+1
23	Oxide is amphoteric (1); Zinc or aluminium (1).	1+1
24	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$ (1); Products: carbon dioxide and water (1).	1+1
25	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
26	Motor neuron or effector (1); Sensory path is intact (1).	1+1

SECTION — C (Short Answer) — Answers

6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Equation already balanced: $PbO + C \longrightarrow Pb + CO$ (1); PbO is the oxidising agent (1); C is the reducing agent (1).	1+1+1
28	pH 10 and 13 turn it pink (1); pH 1 and 4 neutralise a base (1); Acidity order: pH 1 > 4 > 7 > 10 > 13 (1).	1+1+1
29	ZnO is the oxidising agent and C is the reducing agent (1); Redox, reduction of metal oxide with carbon (1); Zinc is moderately reactive, so carbon can reduce its oxide (1).	1+1+1
30	$CH_3-CH=CH_2$ (1); Propene is unsaturated (a double bond); propane is saturated (1); Propene undergoes addition reactions; propane undergoes substitution (1).	1+1+1
31	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
	OR: Tip makes auxin (1); Auxin goes to the shaded side (1); No tip: no auxin, no bending (1).	1+1+1
32	Divisions in 2 hours = 8 (1); Number = 2^8 = 256 (1); Binary fission (asexual) (1).	1+1+1
	OR: Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Reddish brown (the solid is Fe_2O_3) (1). (ii) Thermal decomposition (1). (iii) $2\,FeSO_4 \xrightarrow{heat} Fe_2O_3 + SO_2 + SO_3$ (1). (iv) Sulphur dioxide and sulphur trioxide (1); endothermic, because heat has to be supplied continuously for the reaction to proceed (1).	1+1+1+2
34	(i) Sample Q (pH = 7) (1). (ii) Sample S (highest pH) (1). (iii) R and S (pH greater than 7) (1). (iv) Quick lime / slaked lime / chalk (any one) (1), because it is basic and neutralises the excess acid in the soil (1).	1+1+1+2
35	Order: A > B > C > D (1); A: electrolysis of molten salt (very reactive) (1); D: occurs free or by heating its ore (1); Example set: Mg, Zn, Cu, Ag (1); Reason: a more reactive metal displaces a less reactive one (1).	1+1+1+1+1
36	Dehydration: conc. H_2SO_4, 443 K (1); Oxidation: alkaline $KMnO_4$ (1); $KMnO_4$ is oxidising agent (1); Esterification: conc. H_2SO_4 catalyst (1); Equations of each step (1).	1+1+1+1+1
	OR: Aim: to show CO_2 is needed for photosynthesis (1); KOH absorbs CO_2 (1); Leaf with KOH: no starch (1); Leaf with water: starch present (1); Control: leaf in water flask (1).	1+1+1+1+1
37	Aim: role of tip in phototropism (1); Take three seedlings: normal, tip cut, tip covered with foil (1); Expose to light from one side (1); Observation: only normal seedling bends (1); Conclusion: tip senses light and makes auxin (1).	1+1+1+1+1
	OR: Asexual offspring are nearly identical clones (1); DNA copying errors give small variations only (1); Sexual reproduction mixes DNA of two parents (1); Variations help some survive changes in the environment (1); Example: cross-pollination in plants gives varied seeds (1).	1+1+1+1+1
38	F1: TtRr, tall round (1); F2 ratio 9 : 3 : 3 : 1 (1); Tall round 900, tall wrinkled 300 (1); Dwarf round 300, dwarf wrinkled 100 (1); Dwarf round genotypes ttRR and ttRr (1).	1+1+1+1+1
	OR: Concave: 1/v = -1/20 + 1/30 = -1/60, v = -60 cm (1); m = -2, size 8 cm, real, inverted (1); Convex: 1/v = 1/20 + 1/30 = 1/12, v = +12 cm (1); m = +0.4, size 1.6 cm (1); Convex: virtual, erect, diminished (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 50	1/4 of 200 = 50 (Rryy).
2	**(b)** 11.25 cm	1/f = -1/45 - 1/15 = -4/45, so f = -11.25 cm.
3	**(d)** -25 cm	f = -(far point) for objects at infinity.
4	**(c)** 6 ohm	R2 = rho 2L / 4A = R / 2.
5	**(b)** 90 degrees	Perpendicular is the maximum.
6	**(a)** 5000 J	5 x 10 x 10 x 10.
7	**(b)** copper	CuO is reduced to Cu by hydrogen.
8	**(a)** plaster of Paris	It sets into gypsum.
9	**(d)** CuO and CO_2	Calcination: $CuCO_3 \longrightarrow CuO + CO_2$.
10	**(d)** ethene	C_2H_4 = 28.
11	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
12	**(c)** adrenaline	Fight or flight.
13	**(d)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
14	**(d)** 1 : 1 : 1 : 1	Four gamete types x one.
15	**(b)** convex, f = 12 cm	Object at 2F.
16	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Turgid guard cells bulge apart and open the pore; shrunken ones close it.
18	**(a)** Both A and R are true and R is the correct explanation of A.	The short reflex arc through the spinal cord saves time, which makes the response quick.
19	**(a)** Both A and R are true and R is the correct explanation of A.	With a single parent and no gamete fusion there is no mixing of DNA, so the offspring are almost identical clones.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Every F1 plant is Tt; the dominant allele T masks t, so R explains A.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Regeneration (1); It involves growth of a whole body from a body part, not an outgrowth (1).	1+1
22	TT fraction = 1/4 (1); 60 plants (1).	1+1
23	m = -1/3 = -v/u, v = -15, so u = -45 cm (1); 1/f = -1/15 - 1/45 = -4/45, f = -11.25 cm (1).	1+1
24	Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1).	1+1
25	I = 4.8×10^{-3} A (1); n = 4.8×10^{-3} / 1.6×10^{-19} = 3×10^{16} (1).	1+1
26	Force is perpendicular to the velocity (1); No work is done, so the speed is constant (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	First case: u = -x, v = -3x, so 1/f = -1/(3x) - 1/x = -4/(3x), |f| = 3x/4 (1); Second case: same size means the object is at C, so x + 10 = 2|f| = 3x/2 (1); x = 20 cm, so f = -15 cm (1).	1+1+1
28	1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1).	1+1+1
29	Atoms vibrate more, increasing collisions with electrons (1); Resistance increases (1); Current decreases (I = V / R) (1).	1+1+1
30	Force arises from the component of the field perpendicular to the wire (1); It is maximum when the wire is perpendicular (1); It is zero when the wire is parallel (1).	1+1+1

31	Small fish: 0.2 ppm (1); Big fish: 2 ppm (1); Kingfisher: 20 ppm (1).	1+1+1
	OR: Basic copper carbonate (1); Copper reacts with moisture, CO_2 and oxygen (1); Prevent by lacquer/paint or keeping dry (1).	1+1+1
32	X is $CaCO_3$, Y is CaO (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1+1
	OR: Salt water increases electrical conductivity (1); Faster electrochemical corrosion (1); Rust formation speeds up (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Copper (1). (ii) Hydrogen; it burns with a pop sound with a burning matchstick (1). (iii) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1). (iv) Mg > Zn > Fe > Cu (1); the faster the bubbling, the more reactive the metal is (1).	1+1+1+2
34	(i) C_5H_{12} (pentane) (1). (ii) Ethyne, C_2H_2 (1). (iii) 14 u (a CH_2 unit) (1). (iv) Propane C_3H_8 has $3n + 1 = 10$ covalent bonds (1); all carbon-carbon bonds are single bonds, so each carbon holds the maximum number of hydrogen atoms (saturated) (1).	1+1+1+2
35	Fleming's left-hand rule (1); Forefinger north, middle finger east (1); Thumb points upward (1); (a) Force becomes downward (1); (b) Both reversed: force upward as before (1).	1+1+1+1+1
36	Zooplankton: 200,000 J (1); Small fish: 20,000 J (1); Large fish: 2000 J; birds: 200 J (1); Energy lost at the first transfer = 1,800,000 J (1); The first transfer has the largest loss (1).	1+1+1+1+1
	OR: Green to white (water lost), then reddish brown residue (1); $2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ (1); Gases: SO_2 and SO_3 (1); Smell of burning sulphur (1); Thermal decomposition (1).	1+1+1+1+1
37	Fixed H2O molecules per formula unit (1); $CuSO_4 \cdot 5\,H_2O$: blue to white (1); $FeSO_4 \cdot 7\,H_2O$: green to white then brown (1); $Na_2CO_3 \cdot 10\,H_2O$ loses water to give a white powder (1); Test: add water to the white residue - blue colour for $CuSO_4$ (1).	1+1+1+1+1
	OR: (a) Reacts with air and water (1); (b) Malleable, forms Al_2O_3 layer (1); (c) Pure gold is soft (1); (d) Like ions repel when layers shift (1); Each reason relates to property (1).	1+1+1+1+1
38	Butane and isobutane (1); But-1-ene and but-2-ene (1); Isomers: same formula, different structure (1); Homologues: differ by CH_2 (1); Examples (1).	1+1+1+1+1
	OR: Aerobic: mitochondria, CO_2 + H_2O (1); Yeast: ethanol + CO_2 (1); Muscle: lactic acid (1); Only 2 ATP in anaerobic (1); Complete oxidation yields about 30 ATP (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Heat could cause splashing of acid.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Oxide layer protects the metal.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Incomplete combustion gives soot.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Valves prevent backflow.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but not an explanation of neuron direction itself.
6	**(d)** A is false but R is true.	A is false: most organisms cannot reproduce by regeneration; R is true.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Refracted rays only appear to meet when extended backwards.
9	**(c)** A is true but R is false.	A is true; R is false: the image forms in front of the retina.
10	**(c)** A is true but R is false.	A is true; R is false: copper has low resistivity.
11	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Decomposition recycles nutrients.
13	**(b)** 16 g	40 - 24 = 16 g.
14	**(a)** 25 mL	Ratio 1 : 2.
15	**(d)** S	Each displaces the next, so S is lowest.
16	**(c)** 6	Given in the equation.
17	**(a)** Both A and R are true and R is the correct explanation of A.	The wide field of view of the diminished, erect image lets the driver see more traffic behind.
18	**(a)** Both A and R are true and R is the correct explanation of A.	In myopia the image forms in front of the retina; the diverging lens shifts it back onto the retina.
19	**(a)** Both A and R are true and R is the correct explanation of A.	A low-resistance meter in series does not change the current it is measuring.
20	**(a)** Both A and R are true and R is the correct explanation of A.	If two lines crossed, there would be two directions of the field at the point of intersection.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Grass = 50 x 100 = 5000 J (1); Snake = 5 J (1).	1+1
22	Gases NO_2 and O_2 escape (1); Only solid PbO remains; the total mass is still conserved (1).	1+1
23	Plaster of Paris 145 g (1); Water lost 27 g (1).	1+1
24	Reduction of oxide is easier by carbon (1); So sulphides/carbonates are first converted to oxides (1).	1+1
25	$3n + 1 = 22$, n = 7 (1); C_7H_{16} (1).	1+1
26	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	n-butane: 3 C-C, 10 C-H (1); isobutane: 3 C-C, 10 C-H (1); Same number of bonds; isomers (1).	1+1+1

28	Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
29	Motor nerve carries signal to muscle (1); Effector is the muscle (1); Sensory part not needed (1).	1+1+1
30	Plants have meristematic cells for new growth (1); Humans have complex organ systems (1); Human body parts cannot make a whole body (1).	1+1+1
31	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
	OR: m = -5, v = -5u (1); 1/12 = 1/(5|u|) - ... with u negative: 1/12 = 1/(5|u|) + 1/|u| = 6/(5|u|) (1); |u| = 14.4 cm, v = 72 cm (1).	1+1+1
32	Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1).	1+1+1
	OR: P = 300 W (1); I = 300 / 220 = 1.36 A (1); R = 220 / 1.36 = 161.3 ohm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Salivary amylase (1). (ii) Hydrochloric acid (1). (iii) Small intestine (1). (iv) Bile emulsifies fats, breaking large globules into smaller ones for the action of lipase (1); the pancreatic enzymes act only in an alkaline medium, so the acidic food from the stomach is made alkaline by bile (1).	1+1+1+2
34	(i) Reflex action (1). (ii) Motor neuron (1). (iii) Spinal cord (1). (iv) A reflex is quick, involuntary and does not need thinking (1); writing is voluntary, controlled by the cerebrum and slower (1).	1+1+1+2
35	First two plants: dwarf offspring means each has t (1); 1/4 dwarf shows both are Tt (1); Third plant crossed with tt gives all tall (1); So it must be TT (1); If it were Tt, half would be dwarf (1).	1+1+1+1+1
36	Now 1/v = -1/20 + 1/30 = -1/60, v = -60 cm (1); For image at infinity the object must be at F, 20 cm (1); Move 10 cm towards the mirror (1); Reflected beam is parallel (1); Used in torches and headlights (1).	1+1+1+1+1
	OR: u = infinity, v = -2 m (1); f = -2 m (1); P = -0.5 D (1); Image at the far point, 2 m (1); It is virtual and erect for the lens (1).	1+1+1+1+1
37	20 ohm: P = 14400 / 20 = 720 W (1); 30 ohm: P = 480 W (1); Series 50 ohm: 288 W (1); Parallel 12 ohm: 1200 W (1); Parallel gives the highest power (1).	1+1+1+1+1
	OR: Both have similar field patterns (1); Lines go from N to S outside (1); The solenoid's field can be switched on or off (1); A soft iron core makes it a strong electromagnet (1); Scrap is lifted while the current flows and dropped when switched off (1).	1+1+1+1+1
38	Pesticide kills insects and remains in their bodies (1); Birds that eat insects accumulate it (1); Biomagnification poisons the birds (1); Alternative: biological control with natural predators (1); Alternative: crop rotation or organic farming (1).	1+1+1+1+1
	OR: Zinc: bubbles of hydrogen (1); $\mathrm{Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2}$ (1); Copper: no reaction (1); Zinc is above hydrogen in the reactivity series (1); Copper is below hydrogen so cannot displace it (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.